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Regel:

$$f(x) = \frac{g(x)}{h(x)}$$

$$\Rightarrow f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2}$$

$$f(x) = \frac{x^3 - 1}{x^2 + x}$$

$$\Rightarrow f'(x) = \frac{(3x^2)(x^2 + x) - (x^3 - 1)(2x + 1)}{(x^2 + x)^2} = \frac{(3x^4 + 3x^3) - (2x^4 + x^3 - 2x - 1)}{x^4 + 2x^3 + x^2}$$

$$= \frac{x^4 + 2x^3 + 2x + 1}{x^4 + 2x^3 + x^2}$$

References